



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

REVIEW TEST

02-01-2022

JEE(Main + Advanced) : NURTURE COURSE [PHASE : TOPS, I(A), I, I(B), II(A), II & LIVE-I, I(B), II(A)]

ANSWER KEY**PAPER-1****PART-1 : PHYSICS**

SECTION-I(i)	Q.	1	2	3	4	5	6
	A.	D	D	A	B	A	B
SECTION-I(ii)	Q.	7	8	9	10	11	12
	A.	A,B	B,D	B,C	A	A,B,C,D	A,B,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	13.46	13.00	22.00	4.00	2.50	1.66 to 1.67

PART-2 : CHEMISTRY

SECTION-I(i)	Q.	1	2	3	4	5	6
	A.	D	C	D	C	C	B
SECTION-I(ii)	Q.	7	8	9	10	11	12
	A.	A,C,D	A,B,D	C	C,D	A,B	B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	810.00	90.00	1.00	2.00	5.00	8.00

PART-3 : MATHEMATICS

SECTION-I(i)	Q.	1	2	3	4	5	6
	A.	C	B	C	A	A	C
SECTION-I(ii)	Q.	7	8	9	10	11	12
	A.	A,C	A,B,C,D	A,B,C	B,C,D	B,C	B,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	8.00	4.00	0.75	0.50	4.50	5.00

HINT – SHEET**PART-1 : PHYSICS****SECTION-I(i)**1. **Ans (D)**

$$\rho 2AV = 2 \times (\rho AV')$$

$$F = \rho 2AV^2 + \left[\rho A(V)^2 \right] 2 \times \frac{1}{2} = 3\rho AV^2$$

$$\mu(\rho 2A\ell + \rho A\ell + \rho A\ell)g = 3\rho AV^2$$

$$\mu = \frac{3}{4} \frac{V^2}{g\ell}$$

2. **Ans (D)**

Total work done = change in kinetic energy

$$= \frac{1}{2} mv_2^2 - \frac{1}{2} mv_1^2$$

$$= \frac{1}{2} (2) (4^2 - 2^2) = 12J$$

3. **Ans (A)**

$$F_x = -\frac{dU}{dx} = -2, F_y = -\frac{dU}{dy} = -4y$$

$$2 - N_x = 2; N_z = 0.3 \times 10 = 3$$

$$N = 3$$

4. **Ans (B)**

$$-Ndt = m\omega_2 \frac{L}{2} - mg\omega_1 \frac{L}{2}$$

$$N \frac{L}{2} dt = \frac{mL}{12} \omega_2 + \frac{mL^2}{12} \omega_1$$

$$Ndt = \frac{mL}{6} [\omega_2 + \omega_1]$$

$$\omega_2 = \frac{\omega_1}{2}$$

$$\int Ndt = \frac{ML\omega_1}{4}$$

5. **Ans (A)**

$$P_0 h = P' (h - x)$$

$$2h + h = P' + x$$

$$3h - x = P'$$

$$P' = \frac{P_0 h}{(h - x)}$$

$$(3h - x)(h - x) = h^2$$

6. **Ans (B)**

$$V = Ah$$

$$\frac{\Delta V}{V} = \frac{\Delta A}{A} + \frac{\Delta h}{h}$$

$$\gamma \ell \Delta T - 2\alpha_g \Delta T = \frac{\Delta h}{h}$$

PART-1 : PHYSICS

SECTION-I(ii)

8. **Ans (B,D)**

Let us assume ball land on ground

$$-36 = 3t - \frac{1}{2} 10t^2 \Rightarrow t = 3$$

S_x in $t = 3$ sec $\Rightarrow$ -ve (ball lands on tower)

$$\Rightarrow S_x = 0 = 4xt - \frac{1}{2} \times 10 \times t^2 \Rightarrow t = \frac{8}{5}$$

$$\Rightarrow S_y = 3 \times \frac{8}{5} - \frac{1}{2} \times 10 \left(\frac{64}{25} \right) = -8\text{m}$$

$\vec{V} \cdot \vec{a} = 0$ at minimum velocity

$$V_x a_x + V_y a_y = 0$$

$$(4 - 10t) \times (-10) + (3 - 10t)(-10) = 0$$

$$7 = 20t$$

$$\frac{7}{20} = t$$

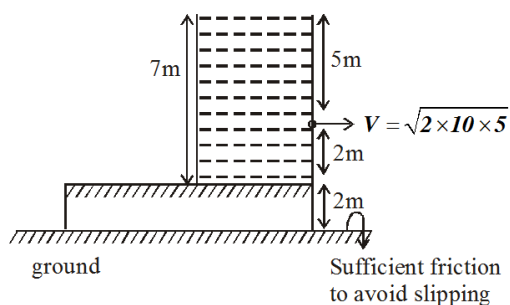
$$t = 0.35 \text{ sec}$$

9. **Ans (B,C)**

$$F = \rho A V^2$$

$$= 1000 \times 10^{-4} \times 2 \times 10 \times 5 = 10 \text{ N}$$

$$4 = \frac{1}{2} \times 10 \times t^2; t = \sqrt{\frac{4}{5}}$$



$$\text{Range} = \sqrt{2 \times 10 \times 5} \times \sqrt{\frac{4}{5}} = 4\sqrt{5}$$

10. **Ans (A)**

$$\mu N \Rightarrow \delta = \frac{Mg}{2\ell^2} (R^2 + \ell^2)$$

$$\frac{\mu MRg}{\ell}; \mu \geq \frac{(R^2 + \ell^2)}{2\ell R}$$

$$\frac{\cos \theta}{2} = \frac{\ell}{\sqrt{R^2 + \ell^2}}$$

$$T \sin \theta = N$$

$$N = mg \tan \frac{\theta}{2} = \frac{MgR}{\ell}$$

$$T = \delta, T \cos \theta + \delta = Mg$$

$$T = \frac{Mg}{\frac{2 \cos^2 \theta}{2}} = \frac{Mg}{2\ell^2} (R^2 + \ell^2)$$

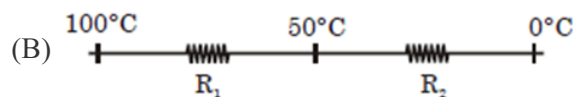
11. **Ans (A,B,C,D)**

$$\text{Thermal resistance } R = \int \frac{dx}{kA}$$

$$R = \int_0^x \frac{dx}{\frac{3}{2} \pi (x+1)^3} = \frac{1}{6} \left[1 - \frac{1}{(x+1)^2} \right]$$

$$R \text{ of complete rod} = \frac{1}{6} \left[1 - \frac{1}{3^2} \right] = \frac{4}{27}$$

$$(A) \frac{d\theta}{dt} = \frac{100}{\frac{4}{27}} = \frac{2700}{4} = 675 \text{ watt}$$



$$R_1 = R_2 \frac{1}{6} \left[1 - \frac{1}{(x+1)^2} \right] = \frac{1}{6} \left[\frac{1}{(x+1)^2} - \frac{1}{9} \right]$$

$$\Rightarrow x \approx 34 \text{ cm}$$

$$(C) x = 1\text{m}$$

$$R = \frac{1}{6} \left[1 - \frac{1}{4} \right] = \frac{1}{8} \Rightarrow 100 - T = \frac{1}{8} \times 625$$

$$\Rightarrow T = 15.6^\circ\text{C}$$

$$(D) \frac{R(x=0 \text{ to } x=1)}{R(x=1 \text{ to } x=2)} = \frac{1 - \frac{1}{4}}{\frac{1}{4} - \frac{1}{9}} = \frac{27}{5}$$

12. Ans (A,B,D)

Figure-1 shows condition of just spilling out of liquid, when tube is accelerated horizontally.

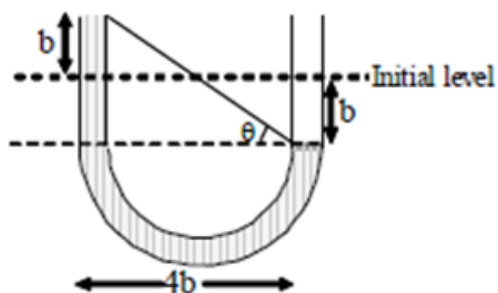


Figure 1

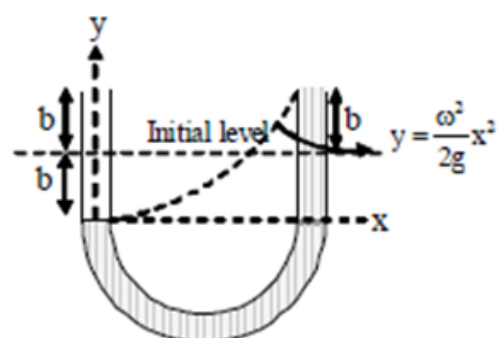


Figure 2

$$\tan \theta = \frac{2b}{4b} = \frac{1}{2} \text{ Also, } \tan \theta = \frac{a}{g} \text{ [a = acceleration of U tube]} \Rightarrow a = \frac{g}{2}$$

Figure-2 Shows condition of just spilling out of water its rotated about left arm

At $x = 4b$; $y = 2b$;

Putting value in $y = \frac{\omega^2}{2g} x^2$ we have

$$2b = \frac{\omega^2}{2g} (4b)^2 \Rightarrow \omega = \sqrt{\frac{g}{4b}}$$

Similarly for BB'

$$y = \frac{\omega^2}{2g} x^2 \Rightarrow b = \frac{\omega^2}{2g} (2b)^2 \Rightarrow \omega = \sqrt{\frac{g}{2b}}$$

PART-1 : PHYSICS

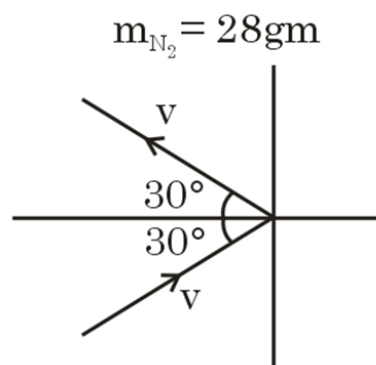
SECTION-II

1. Ans (13.46)

$$A = 1 \text{ cm}^3 = 10^{-6} \text{ m}^3$$

$$n = 0.9 \times 10^{19}$$

$$v = 400 \text{ m/s}$$



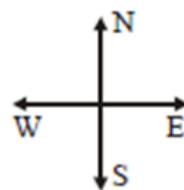
$$P = 2mnv^2 \cos^2 \theta$$

$$= \frac{28}{6.02 \times 10^{26}} \times 6.02 \times 10^{13} \times 16 \times 10^4 \times \frac{3}{2}$$

$$= 1344 \times 10^{-9} = \frac{13.44}{2} \times 10^{-7}$$

2. Ans (13.00)

$$V_B = 8\hat{i}, V_{SB} = 12\hat{j} - 2\hat{k}$$



$$V_{HS} = -\hat{i} + 5\hat{k}$$

$$\vec{V}_{HB} = \vec{V}_H - \vec{V}_B = \vec{V}_{HS} + \vec{V}_{SB} = -4\hat{i} + 12\hat{j} + 3\hat{k}$$

$$|\vec{V}_{HB}| = \sqrt{(4)^2 + (12)^2 + (3)^2} = 13 \text{ m/sec}$$

3. Ans (22.00)

$$\text{Velocity at C} = \sqrt{2gr}$$

$$a_{\text{net}} = \sqrt{g^2 + (2g)^2} = \sqrt{5}g$$

4. **Ans (4.00)**

By conservation of angular momentum about point P

Let speed of bug w.r.t. ground is v_0

$$L_i = L_f = 0 \quad m_{\text{ring}} = 2m_{\text{bug}}$$

$$L_{\text{ring}} = L_{\text{bug}} \quad m_{\text{bug}} = m$$

$$4mR^2\omega = m2R v_0$$

$$\omega = \frac{v_0}{2R}$$

By relative motion $v_0 + 2R\omega = v$

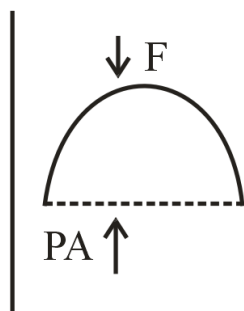
$$v_0 + v_0 = v$$

$$v_0 = \frac{v}{2} = \frac{8}{2} = 4 \text{ m/s}$$

5. **Ans (2.50)**

$$2a = v \frac{dm}{dt} = 5 \times 1$$

6. **Ans (1.66 to 1.67)**



$$PA - F = F_B = \frac{2\pi}{3} r^3 \rho_1 g$$

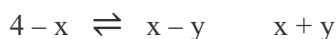
$$(P_0 + \rho_1 gh) \pi r^2 - F = \frac{2\pi}{3} r^3 \rho_1 g$$

$$F = P_0 \pi r^2 + \left(h - \frac{2}{3} r \right) \pi r^2 \rho_1 g$$

PART-2 : CHEMISTRY

SECTION-I(i)

1. **Ans (D)**



$$2.5 = \frac{(x+y)(y-x)}{4-x} \Rightarrow 4-x = x-y$$

$$\text{and } \text{O}_2 = x + y \Rightarrow 2.5$$

$$\text{Hence, } [\text{N}_2\text{O}] = y = \frac{1}{3} \text{ M.}$$

PART-2 : CHEMISTRY

SECTION-II

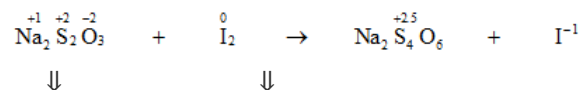
1. **Ans (810.00)**

$$P_A - 90 = P_B \quad P_B + 90 = 760 \text{ mm of Hg}$$

P_A & P_B are in mm of Hg

$$P_A = 90 + 720 = 810 \text{ mm of Hg}$$

2. **Ans (90.00)**



$$(n_f = 1) \quad (n_f = 2)$$

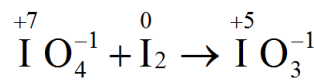
$$\therefore \text{Meq. of } \text{Na}_2\text{S}_2\text{O}_3 = \text{meq. of } \text{I}_2$$

$$\Rightarrow M \times V_{\text{mL}} \times n_f = \text{mmoles} \times n_f$$

$$\therefore \text{m moles of } \text{I}_2 = \frac{0.1 \times 10000 \times 1}{2}$$

$$\therefore \text{m moles of } \text{I}_2 = 50 \rightarrow \text{from 1}^{\text{st}} \text{ reaction}$$

Similarly for 2nd reaction



$$n_f = 2 \quad n_f = 10$$

$$\Rightarrow \text{meq. of } \text{IO}_4^{-1} = \text{meq. of } \text{I}_2$$

$$M \times V_{\text{mL}} \times n_f = (\text{m moles}) \times n_f$$

$$\therefore \text{m moles of } \text{I}_2 = \frac{0.2 \times 1000 \times 2}{10}$$

$$\therefore \text{m moles of } \text{I}_2 = 40 \text{ from 2}^{\text{nd}} \text{ reaction}$$

$$\text{total m moles of } \text{I}_2 = 50 + 40 = 90 \text{ m moles}$$

PART-3 : MATHEMATICS

SECTION-I(i)

1. **Ans (C)**

For sum of coefficients, put $x = 1$

$$\therefore \text{sum of coeff.} = (-1 + (a - 2b + c))^{2015}$$

$$= (-1)^{2015} = -1$$

2. **Ans (B)**

$$\frac{(2-x^2)(x-3)^3}{(x+1)(x^2-3x-4)} \leq 0$$

$$\frac{(x+\sqrt{2})(x+\sqrt{2})(x-3)^3}{(x+1)(x-4)(x+1)} \leq 0$$

$$\frac{(x+\sqrt{2})(x+\sqrt{2})(x-3)^3}{(x+1)^2(x-4)} \leq 0$$

$$(-\sqrt{2}, -1) \cup (-1, \sqrt{2}) \cup [3, 4)$$

$$x = 0, 1, 3$$

3. **Ans (C)**

Equation of common tangent is

$$y = mx \pm \sqrt{4m^2 + 1} \text{ \& } y = nx \pm \sqrt{m^2 + 4}$$

$$\Rightarrow y + x = \sqrt{5}$$

$$x + y - \sqrt{5} = 0$$

$$2a = \frac{3 - \sqrt{5}}{\sqrt{2}} \Rightarrow 4a = \sqrt{2}(3 - \sqrt{5})$$

4. **Ans (A)**

$$\sum_{n=1}^{100} a_{3n} - \sum_{n=1}^{100} a_{3n-2} = 100$$

$$\Rightarrow 100(2d) = 100 \Rightarrow d = \frac{1}{2}$$

$$\& \frac{100}{2}(2a_1 + 99.3d) = 100$$

$$\Rightarrow a_1 = -\frac{293}{4}$$

5. **Ans (A)**

$$9^{\log_3(\log_2 x)} = \log_2 x - (\log_2 x)^2 + 1$$

$$(\log_2 x)^2 = (\log_2 x) - (\log_2 x)^2 + 1$$

$$\log_2 x = t$$

$$2t^2 - t - 1 = 0, (2t + 1)(t - 1) = 0$$

$$t = 1 \text{ (only)}$$

$$x = 2$$

6. **Ans (C)**

Do yourself

PART-3 : MATHEMATICS

SECTION-I(ii)

7. **Ans (A,C)**

$$I + F = (15 + \sqrt{224})^n = {}^nC_0(15)^n$$

$$+ {}^nC_1(15)^{n-1}(\sqrt{224})^1 + \dots + {}^nC_n(15)^0(\sqrt{224})^n$$

$$\text{Let } F' = (15 - \sqrt{224})^n$$

$$= {}^nC_0(15)^n - {}^nC_1(15)^{n-1}(\sqrt{224})^1 + \dots$$

$$\Rightarrow I + F + F' = \text{even}$$

$$\Rightarrow 0 < F + F' < 2 \Rightarrow F + F' = 1$$

$$\Rightarrow F' = 1 - F$$

$$\Rightarrow I = \text{odd and } x(1 - F)$$

$$= (15 + \sqrt{224})^n (15 - \sqrt{224})^n = 1$$

8. **Ans (A,B,C,D)**

$${}^3C_2 \times 2! \times 4! = 3 \times 2 \times 24 = 144$$

9. **Ans (A,B,C)**

a = 2, b = 3, c = 4 now check options

10. **Ans (B,C,D)**

$$x^2 + bx - 1 = 0$$

$$bx^2 + x - 1 = 0$$

let α is common solution

$$\frac{\alpha^2}{-b+1} = \frac{\alpha}{-b+1} = \frac{1}{1-b^2}$$

$$\therefore \alpha^2 = \frac{-b+1}{1-b^2}, \alpha = \frac{-b+1}{1-b^2}$$

$$\Rightarrow \frac{-b+1}{1-b^2} = \left(\frac{-b+1}{1-b^2} \right)^2 \Rightarrow b = 0 \text{ or } 1$$

at b = 0

exactly one common root

at b = 1 both roots are common

11. **Ans (B,C)**

$$(\log_x 2 + \log_y 2 + \log_z 2)^2 = (\log_x 2)^2 + (\log_y 2)^2 + (\log_z 2)^2$$

$$+ 2(\log_x 2 \cdot \log_y 2 + \log_y 2 \cdot \log_z 2 + \log_z 2 \cdot \log_x 2)$$

$$\Rightarrow \log_x 2 \cdot \log_y 2 + \log_y 2 \cdot \log_z 2 + \log_z 2 \cdot \log_x 2 = 0$$

$$\Rightarrow \frac{\log_2 x + \log_2 y + \log_2 z}{\log_2 x \cdot \log_2 y \cdot \log_2 z} = 0 \Rightarrow \log_2 (xyz) = 0$$

$$\Rightarrow xyz = 1$$

12. Ans (B,D)

$$\text{given } \frac{\tan 3x - \tan 2x}{1 + \tan 3x \tan 2x} = 1$$

$$\tan x = 1$$

$$x = n\pi + \frac{\pi}{4}$$

both for this value of x , $\tan 2x$ is not defined.

PART-3 : MATHEMATICS

SECTION-II

1. Ans (8.00)

$$\text{coeff of } x^2 = \text{coeff of } x^3 \Rightarrow a = \frac{9}{7}$$

$$\text{Now, } f\left(\frac{9}{7}\right) < 0 \Rightarrow p < -8$$

$$\text{Hence, } K = 8$$

2. Ans (4.00)

$$\frac{2 \sin 30^\circ \cdot \cos 2^\circ - \frac{\sqrt{3}}{2}}{\frac{1}{2}(\cos 2^\circ - \cos 30^\circ) \cdot \frac{1}{2}} = 4$$

3. Ans (0.75)

Use AM $\geq$ GM

$$\frac{\left(\frac{4}{\alpha\beta} + \frac{3}{\beta\gamma} + \frac{2}{\gamma\alpha}\right)}{3} \geq \left(\frac{4}{\alpha\beta} \cdot \frac{3}{\beta\gamma} \cdot \frac{2}{\gamma\alpha}\right)^{\frac{1}{3}}$$

$$\frac{4}{\alpha\beta} + \frac{3}{\beta\gamma} + \frac{2}{\gamma\alpha} \geq \frac{3}{4}$$

4. Ans (0.50)

$$P(x) = (x - 11)(x - 99)Q(x) + ax + b$$

$$P(11) = 11a + b = 53 \quad \dots(1)$$

$$P(99) = 99a + b = -35 \quad \dots(2)$$

$$-88a = 88 \therefore a = -1, b = 64$$

$$r(x) = -x + 64$$

$$r(63) = 1 \text{ or } \frac{r(63)}{2} = \frac{1}{2} = 0.50$$

5. Ans (4.50)

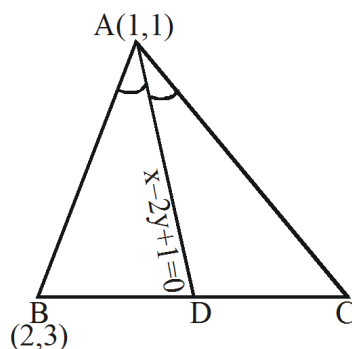


Image of B with respect to AD lies on AC and

$$\text{image of } B(2,3) \text{ is } \left(\frac{16}{5}, \frac{3}{5}\right)$$

$$\therefore \text{slope of } AC = \frac{2/5}{-11/5} = -\frac{2}{11}$$

$$\Rightarrow |m| = \frac{2}{11} = \frac{a}{b} \Rightarrow b - a = 9$$

$$\frac{b - a}{2} = 4.50$$

6. Ans (5.00)

$$\text{Let } S = 1 \cdot ({}^{20}C_0)^2 + \dots + 11 \cdot ({}^{20}C_{20})^2$$

$$S = 11 \cdot ({}^{20}C_{20})^2 + \dots + 1 \cdot ({}^{20}C_0)^2$$

$$2S = 12 \left[({}^{20}C_0)^2 + ({}^{20}C_2)^2 + \dots + ({}^{20}C_{20})^2 \right]$$

$$S = 6 \left[\frac{({}^{20}C_0)^2 + ({}^{20}C_1)^2 + \dots + ({}^{20}C_{20})^2}{2} \right]$$

$$S = 3[{}^{40}C_{20} + {}^{20}C_{10}]$$



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REVIEW TEST

02-01-2022

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ANSWER KEY**PAPER-2****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6
	A.	B,C,D	A,B,C	A,C	A,B,C	C,D	A,B,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	5.25	0.73	6.00	2.5	6.00	4.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	2	3	3	4	4	2

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6
	A.	A,B,D	A,C,D	C,D	A,B,C,D	B,C	A,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	6.00	3.00	4.00	6.00	2.50	1.41
SECTION-III	Q.	1	2	3	4	5	6
	A.	8	3	5	2	7	4

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	B	B,D	A,B,C	A,C,D	C	A,B,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	10.00	2.00	3.00	21.00	0.00	12.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	6	2	1	2	7	6

HINT – SHEET**PART-1 : PHYSICS****SECTION-I**2. **Ans (A,B,C)**

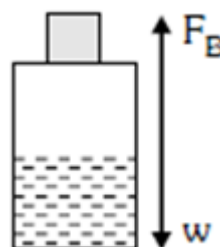
Acceleration will be less than $g \left(\frac{\rho - \sigma}{\rho} \right)$

Time will depend in density ρ . Viscous force may be maximum or minimum both are possible

3. **Ans (A,C)**

$$F_B = w$$

$$(\pi r^2 \ell + V) dg = V (2d)g + \pi r^2 \left(\frac{2\ell}{3} \right) dg$$



$$V = \frac{\pi r^2 \ell}{3}$$

$$P_{\text{air}} = P_0 + dg\ell \left(h + \frac{\ell}{3} \right)$$

4. **Ans (A,B,C)**

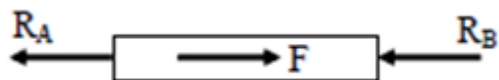
(A) Product of PV is maximum

(B) $P-V = nRT$

(C) At A & C temperature is same

(D) Can not predict must be hyperbola.

5. **Ans (C,D)**



$$\Rightarrow R_A + R_B = f$$

and elongation in both parts must be same

$$\frac{R_B \left(\frac{3L}{4} \right)}{AY} = \frac{R_A \left(\frac{L}{4} \right)}{AY} \Rightarrow 3R_B = R_A$$

6. **Ans (A,B,D)**

As relative velocity after collision ($t = 5$ sec)

becomes zero, therefore $e = 0$

Now,

$$\Delta K_{\text{loss}} = \frac{1}{2} \times \frac{m_1 m_2}{m_1 + m_2} \times (u_1 - u_2)^2 (1 - e^2)$$

$$= \frac{1}{2} \times \frac{2 \times 2}{2 + 2} \times (4)^2 \times (1 - 0) = 8J$$

$$J = mv_1 - 0 = 2 \times 2 = 4 \text{ N-s}$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (5.25)**

$$W_{AB} = -2P_0 V_0$$

$$W_{CD} = +13.5P_0 V_0$$

$$W_{EF} = -4P_0 V_0$$

$$W_{GH} = 3P_0 V_0$$

$$W_{BC} = W_{DE} = W_{RQ} = W_{HA} = 0$$

$$\text{Total work done} = 10.5P_0 V_0 = 2NP_0 V_0$$

$$N = 5.25$$

2. **Ans (0.73)**

$$1.5 \text{ cm} - (\alpha_A L_A \Delta T + \alpha_B L_B \Delta T)$$

4. **Ans (2.5)**

Since rate of heat flow remains same in both the cases,

so thermal resistance should be equal

$$\frac{\ln\left(\frac{R_2}{R_1}\right)}{2\pi k \ell}$$

5. **Ans (6.00)**

$$I = \Delta P$$

$$30 = 3(v_f - 0) \Rightarrow v_f = 10 \text{ m/s}$$

$$\Rightarrow v_{\text{cm}} = \frac{3 \times 10 + 2 \times 0}{5} = 6 \text{ m/s}$$

Velocity of centre of mass remains constant because there is no external force on the system

6. **Ans (4.00)**

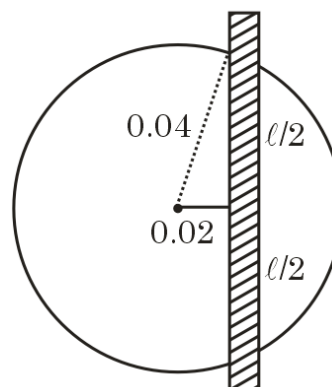
$$a = \frac{50 - 20}{15} = 2 \text{ m/s}^2$$

$$v = \sqrt{2 \times 2 \times 4} = 4 \text{ m/s}$$

PART-1 : PHYSICS

SECTION-III

1. **Ans (2)**



$$\frac{\ell}{2} = \sqrt{(0.04)^2 - (0.02)^2}$$

$$F = 2\gamma \ell$$

$$F = 4\gamma \sqrt{(0.04)^2 - (0.02)^2}$$

2. **Ans (3)**

P-V graph is straight line through origin

$$\therefore P \propto V$$

$$PV^{-1} = \text{constant}$$

polytropic index $m = -1$

$$c = \frac{R}{\gamma - 1} + \frac{R}{-m + 1}$$

$$c = \frac{R}{1.4 - 1} + \frac{R}{1 - (-1)} = xR$$

$$\therefore x = 3$$

3. **Ans (3)**

Friction is not sufficient to prevent slipping

4. **Ans (4)**

By conservation of angular momentum
(About hinge point)

$$(mv_0)2R = \left[\frac{3}{2}mR^2 + m(2R)^2 \right] \omega;$$

$$\omega = \frac{4v_0}{11R} = 4\text{rad/s}$$

5. **Ans (4)**

$$T_1 = \frac{2(2m_1) \cdot m_1}{3m_1}g = \frac{4m_1}{3}g$$

$$T_2 = \frac{2(4m_2)(2m_2)}{6m_2}g = \frac{8}{3}m_2g$$

$$2T_1 \cdot \frac{\ell}{3} = 2T_2 \left(\frac{2\ell}{3} \right)$$

$$T_1 = 2T_2$$

$$\frac{4m_1}{3}g = 2 \times \frac{8}{3}m_2g$$

$$m_1 = 4m_2$$

6. **Ans (2)**

Displacement of mass A is 10 cm

$$m \times 10 = (m + m + 3m) \times x$$

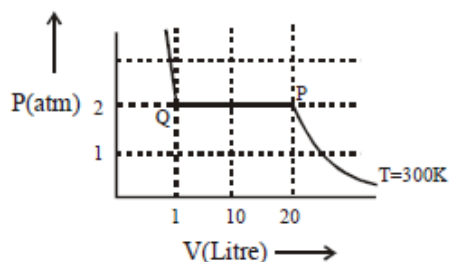
$$10m = 5xm$$

$$x = 2\text{cm}$$

PART-2 : CHEMISTRY

SECTION-I

6. **Ans (A,C,D)**



(A) A real gas condense if external pressure is just greater than vapour pressure
(B,C,D)

At point 'P' whole real gas is in gaseous phase

$$\text{so density of gas} = \frac{1000}{20 \times 1000} = 0.05\text{gm/ml}$$

At point 'Q' whole substance is in liquid phase

$$\text{so density of liquid} = \frac{1000}{1 \times 1000} = 1\text{gm/ml}$$

Note : during condensation at 300K density of gas and liquid remains constant, only amount varies.

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (6.00)**

For $Z=1$

$$\therefore \Delta E = 2.55 = 13.6 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

$$n_1 = 2, n_2 = 4$$

$$\text{maximum no. of lines} = \frac{4(4-1)}{2} = 6$$

2. **Ans (3.00)**



$$t=0 \quad P_0 \quad 0 \quad 0$$

$$t=t_{eq.} \quad P_0(1-\alpha) \quad P_0\alpha \quad \frac{P_0\alpha}{2}$$

$$P = P_0 \left(1 + \frac{\alpha}{2} \right)$$

$$K_P = \frac{(P_0\alpha)^2 \left(\frac{P_0\alpha}{2} \right)}{(P_0(1-\alpha))^2}$$

$$K_P = \frac{P}{(2+\alpha)} \alpha^3$$

$$2+\alpha \simeq 2; \sqrt[3]{\frac{2K_P}{P}} = \alpha$$

PART-2 : CHEMISTRY

SECTION-III

1. **Ans (8)**

$$9a + a + 3.5a = 216$$

$$a = 16$$

Formula is $C_{12}H_{16}N_4$.

2. **Ans (3)**

$$MV = M_1V_1 + M_2V_2$$

$$M = \frac{\frac{20}{11.35} \times 50 + \frac{10}{11.35} \times 200}{1000}$$

$$M = \frac{3}{11.35}$$

$$V = \frac{3}{11.35} \times 11.35$$

$$V = 3$$

PART-3 : MATHEMATICS

SECTION-I

1. **Ans (B)**

$$\begin{aligned} \sum_{r=0}^{10} \frac{{}^{10}C_r}{r+1} \\ &= \sum_{r=0}^{10} \frac{1}{11} {}^{11}C_{r+1} \\ &= \frac{1}{11} [{}^{11}C_1 + {}^{11}C_2 + \dots + {}^{11}C_{11}] \\ &= \frac{1}{11} [{}^{11}C_1 + {}^{11}C_2 + \dots + {}^{11}C_{11} + {}^{11}C_0 - 1] \\ &= \frac{2^{11} - 1}{11} \end{aligned}$$

2. **Ans (B,D)**

$$y = \frac{x^2 + 2x - 11}{x - 3} \Rightarrow x^2 + x(2 - y) + 3y - 11 = 0$$

for real x

$$\Rightarrow D \equiv 4 + y^2 - 4y - 12y + 44 \geq 0$$

$$\Rightarrow y^2 - 16y + 48 \geq 0$$

$$\Rightarrow (y - 12)(y - 4) \geq 0$$

$$\Rightarrow y \in (-\infty, 4] \cup [12, \infty)$$

$$\Rightarrow a = 4 ; b = 12$$

3. **Ans (A,B,C)**

$$\frac{(x-1)^2}{25} - \frac{(y-1)^2}{16} = 1$$

$$\therefore \text{eccentricity} = \sqrt{1 + \frac{16}{25}} = \frac{\sqrt{41}}{5}$$

$$\text{directrix} \Rightarrow x = \pm \frac{25}{\sqrt{41}} + 1$$

$$\text{Foci } X = \pm ae, Y = 0$$

$$\Rightarrow x = \pm \sqrt{41} + 1, y = 1$$

$$\text{Latus rectum} = \frac{2b^2}{a} = \frac{32}{5}$$

4. **Ans (A,C,D)**

The number of ways of placing A_3, A_4 & A_5 so that A_3 comes before A_4 and A_4 comes before A_5 is 8C_3 . There are $5!$ ways of arranging the five other candidates.

$$\text{Hence total arrangements} = {}^8C_3 \cdot 5! = \frac{8!}{6}.$$

5. **Ans (C)**

$${}^nC_1 \cdot 2^1 + {}^nC_2 \cdot 2^2 + {}^nC_3 \cdot 2^3 + \dots + {}^nC_n \cdot 2^n$$

$$(1 + 2)^n = {}^nC_0 + {}^nC_1 \cdot 2^1 + {}^nC_2 \cdot 2^2 + \dots + {}^nC_n \cdot 2^n$$

$$(3^n - 1) = {}^nC_1 \cdot 2^1 + {}^nC_2 \cdot 2^2 + \dots + {}^nC_n \cdot 2^n$$

6. **Ans (A,B,C,D)**

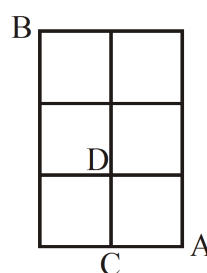
$$2\cos^2 \theta - 2\sqrt{2} \cos \theta - 3 = 0$$

$$\Rightarrow \cos \theta = -\frac{1}{\sqrt{2}} \text{ or } \cos \theta = +\frac{3}{\sqrt{2}} \text{ (rejected)}$$

PART-3 : MATHEMATICS

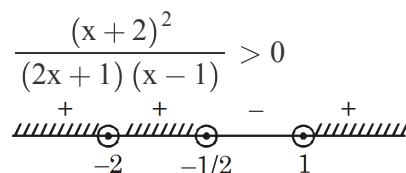
SECTION-II

1. **Ans (10.00)**



$$\text{Total number of ways} = \frac{5!}{3!2!} = 10$$

2. **Ans (2.00)**



$$x \in (-\infty, -2) \cup \left(-2, -\frac{1}{2}\right) \cup (1, \infty)$$

3. **Ans (3.00)**

$$A = \frac{a+b}{2}, G = \sqrt{ab} \text{ \& \ } H = \frac{2ab}{a+b}$$

$$\text{Now } A = 3H$$

$$\Rightarrow \frac{a+b}{2} = \frac{6ab}{a+b} \Rightarrow (a+b)^2 = 12ab$$

$$\Rightarrow \frac{\left(\frac{a+b}{2}\right)^2}{(\sqrt{ab})^2} = 3 \Rightarrow \frac{A^2}{G^2} = 3$$

4. Ans (21.00)

$$\sum_{r=0}^{18} (r+1)!(r^2 + 4(r+1) + 2) = k!$$

$$\sum_{r=0}^{18} (r+1)!(r+2)^2 = \sum_{r=0}^{18} (r+2)!(r+2)$$

$$\text{Let } T_r = ((r+3)! - (r+2)!)$$

$$3! - 2!$$

$$4! - 3!$$

⋮

$$21! - 20!$$

$$\text{Sum} = 21! - 2!$$

$$k! = 21!$$

5. Ans (0.00)

$$\frac{1}{\log_a abc} + \frac{1}{\log_{bc} abc} - \log_a abc + \frac{1}{\log_{bc} a}$$

$$\Rightarrow \log_{abc} a + \log_{abc} bc - \log_a abc + \log_a bc$$

$$\Rightarrow \log_{abc} abc - \left[\log_a \frac{abc}{bc} \right]$$

$$\Rightarrow \log_{abc} abc - \log_a a = 0$$

6. Ans (12.00)

$$\text{tangent is } y = 2x + \frac{3}{2}$$

$$m = 2 \text{ \& } c = \frac{2a}{m} = \frac{3}{2}$$

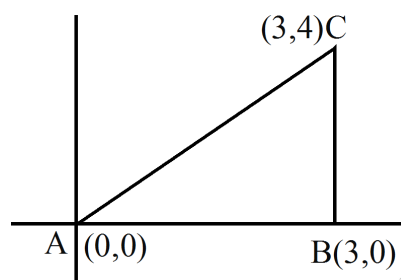
$$\Rightarrow a = \frac{3}{2}$$

$$\text{length of L.R.} = 8a = 12$$

PART-3 : MATHEMATICS

SECTION-III

1. Ans (6)



$$AC = 5 = 2a$$

$$\Rightarrow 2ae = 5 \dots (1)$$

$$\text{Also } AB + BC = 2a$$

$$\Rightarrow 2a = 7$$

$$\Rightarrow a = \frac{7}{2}$$

$$\therefore e = \frac{5}{7} \Rightarrow 1 - \frac{b^2}{a^2} = \frac{25}{49}$$

$$\Rightarrow b = \sqrt{6}$$

$$\text{Area} = \pi ab = \pi \times \frac{7}{2} \times \sqrt{6}$$

$$\therefore P = 6$$

2. Ans (2)

$$\underbrace{\sqrt{(x+7)(x-2)}}_{\downarrow \text{Non-ve}} + \underbrace{|(x+6)(x-2)|}_{\downarrow \text{Non-ve}} = 0$$

$$\text{So, } \sqrt{(x+7)(x-2)} = |(x+6)(x-2)| = 0$$

$$\therefore x = 2$$

3. Ans (1)

$$\alpha + \beta = -1 \quad \alpha\beta = -\lambda^2$$

$$\alpha^2 + \beta^2 + \alpha\beta$$

$$= (\alpha + \beta)^2 - \alpha\beta = 1 + \lambda^2$$

$$\text{Range} = [1, \infty)$$

4. **Ans (2)**

$$a_6 = 4a_4 \Rightarrow ar^5 = 4ar^3 \Rightarrow r^2 = 4 \Rightarrow r = \pm 2$$

 but for increasing, $r = 2$

$$a_9 - a_7 = 192$$

$$\Rightarrow ar^8 - ar^6 = 192$$

$$ar^6(r^2 - 1) = 192$$

$$a2^6(3) = 192$$

$$\Rightarrow a = 1$$

$$\sum_{i=1}^{\infty} \frac{1}{a_i} = \frac{1}{1} + \frac{1}{2} + \frac{1}{4} + \dots + \infty$$

$$= \frac{1}{(1 - \frac{1}{2})} = 2$$

 5. **Ans (7)**

$$|x - 1| + 2 \leq 5$$

$$\Rightarrow |x - 1| \leq 3$$

$$\Rightarrow -3 \leq x - 1 \leq 3$$

$$\Rightarrow -2 \leq x \leq 4$$

 $\therefore$ Integral value of $x = -2, -1, 0, 1, 2, 3, 4$

 6. **Ans (6)**

$$\text{Let } S : x^2 + y^2 - 6x - 10y + \lambda = 0$$

$$\Rightarrow \begin{cases} S_1 < 0 \\ g^2 < c \\ f^2 < c \end{cases} \Rightarrow \begin{cases} 4 + 16 - 12 - 40 + \lambda < 0 \\ 9 < \lambda \\ 25 > \lambda \end{cases}$$

$$\Rightarrow 25 < \lambda < 32$$